1 PolyLog

PolyLog may be defined as the series below

$$\mathrm{Li}_s(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^z}, |z| < 1$$

Or integral

$$\mathrm{Li}_s(z) = \frac{z}{\Gamma(s)} \int_0^{\infty} \frac{x^{s-1}}{\mathrm{e}^x - z} \mathrm{d}x$$

1.1 Prerequisites

A

Let's review the well-known Gamma Function

$$\Gamma(z) = \int_0^{\infty} x^{z-1} \mathrm{e}^{-x} \mathrm{d}x, \Re(z) > 0$$

Integrate by parts we get:

$$\Gamma(z+1) = z\Gamma(z)$$

From the Dominated convergence theorem, Gamma Function could be denoted as

$$\Gamma(z) = \lim_{n \to \infty} \int_0^n (1 - \frac{x}{n})^n x^{z-1} \mathrm{d}x$$

Simplify after recurring IBP, we have Weierstrass's product form:

$$\Gamma(z) = \frac{1}{z} \mathrm{e}^{-\gamma z} \prod_{n=1}^{\infty} ((1 + \frac{z}{n})^{-1} \mathrm{e}^{\frac{z}{n}})$$

Euler's Reflection Formula:

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$$

It's not hard to prove Legendre's Duplication Formula by the trigonometric form of Beta Function:

$$\Gamma(2z) = \frac{2^{2z-1}}{\sqrt{\pi}} \Gamma(z)\Gamma(z + \frac{1}{2})$$

Beta Function:

$$\mathrm{B}(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1} \mathrm{d}x = 2 \int_0^{\frac{\pi}{2}} \sin^{2p-1} x \cos^{2q-1} x \mathrm{d}x, \Re(p) > 0, \Re(q) > 0$$

Which is closely associated with Gamma Function:

$$\mathrm{B}(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

PolyGamma is defined by the nth derivative of the logarithm of Gamma Function:

$$\psi(z) = \frac{\mathrm{d}}{\mathrm{d}z}\log\Gamma(z) = \frac{\Gamma'(z)}{\Gamma(z)}$$

Series representation:

$$\psi(z) = -\gamma - \frac{1}{z} + \sum_{k=1}^{\infty}(\frac{1}{k} - \frac{1}{k+z})$$

$$\psi^{(n)}(z) = (-1)^{n+1}n!\sum_{k=0}^{\infty}\frac{1}{(z+k)^{n+1}}$$

Integral representation:

$$\psi(z) = -\gamma + \int_0^1 \frac{1-x^{z-1}}{1-x}\mathrm{d}x$$

B

Series

$$\zeta(z) = \sum_{n=1}^{\infty}\frac{1}{n^z}, \Re(z) > 1$$

$$\eta(z) = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n^z}, \Re(z) > 1$$

$$\lambda(z) = \sum_{n=1}^{\infty}\frac{1}{(2n-1)^z}, \Re(z) > 0$$

$$\beta(z) = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{(2n-1)^z}, \Re(z) > 0$$

They are commonly applied to integrals

$$\int_0^{\infty}\frac{x^{z-1}}{\mathrm{e}^x - 1}\mathrm{d}x = \Gamma(z)\zeta(z)$$

$$\int_0^{\infty}\frac{x^{z-1}}{\mathrm{e}^x + 1}\mathrm{d}x = \Gamma(z)\eta(z)$$

$$\int_0^{\infty}\frac{x^{z-1}}{\mathrm{e}^x - \mathrm{e}^{-x}}\mathrm{d}x = \Gamma(z)\lambda(z)$$

$$\int_0^{\infty}\frac{x^{z-1}}{\mathrm{e}^x + \mathrm{e}^{-x}}\mathrm{d}x = \Gamma(z)\beta(z)$$

As well as Bernoulli and Euler numbers

$$\zeta(2n) = \frac{(-1)^{n-1}(2\pi)^{2n}}{2(2n)!}B_{2n}$$

$$\beta(2n+1) = \frac{(-1)^n\pi^{2n+1}}{4^{n+1}(2n)!}E_{2n}$$

C

Fourier Series

$$\log\sin x = -\log 2 - \sum_{n=1}^{\infty}\frac{\cos 2nx}{n}$$
$$\log\cos x = -\log 2 - \sum_{n=1}^{\infty}\frac{(-1)^n\cos 2nx}{n}$$

The goal is to compute the Fourier series of $g(x)=\log\sin x$ over $[0,\pi]$, or the Fourier series of $h(x)=\log\sin\frac{x}{2}$ over $[0,2\pi]$, or the Fourier series of $f(x)=\log\cos\frac{x}{2}$ over $[-\pi,\pi]$. Since $f(x)$ is an even function, we have to compute:

$$a_n = \frac{1}{\pi}\int_{-\pi}^{+\pi}\cos(nx)\log\cos\frac{x}{2}\mathrm{d}x = \frac{2}{\pi}\int_{0}^{\pi}\cos(nx)\log\cos\frac{x}{2}\mathrm{d}x$$

for any $n\geq 1$ to be able to state:

$$f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi}f(x)\mathrm{d}x + \sum_{n\geq 1}a_n\cos(nx) = \frac{a_0}{2}+\sum_{n\geq 1}a_n\cos(nx).$$

for any $x\in(-\pi,\pi)$. Integration by parts gives:

$$a_n = \frac{2}{\pi}\left(\frac{1}{n}\sin(nx)\log\cos\frac{x}{2}\Big|_0^{\pi}+\frac{1}{2n}\int_0^{\pi}\sin(nx)\tan\frac{x}{2}\mathrm{d}x\right)$$

or just:

$$a_n = \frac{1}{\pi n}\int_0^{\pi}\frac{\sin(nx)\sin(x/2)}{\cos(x/2)}\mathrm{d}x = -\frac{2}{\pi n}\int_0^{\frac{\pi}{2}}\frac{\cos((2n+1)x)}{\cos x}\mathrm{d}x.$$

Since $\cos((2n+1)x)=2\cos x\cos(2nx)-\cos((2n-1)x)$, we have:

$$\int_0^{\frac{\pi}{2}}\frac{\cos((2n+1)x)}{\cos x}\mathrm{d}x = -\int_0^{\frac{\pi}{2}}\frac{\cos((2n-1)x)}{\cos x}\mathrm{d}x = (-1)^n\frac{\pi}{2}.$$

This gives:

$$\log\cos\frac{x}{2} = \frac{a_0}{2}+\sum_{n\geq 1}\frac{(-1)^{n+1}}{n}\cos(nx)$$

for any $x\in(-\pi,\pi)$. In order to find a_0, we can simply match $f(0)=0$ with the series on the right hand side. Since:

$$\sum_{n\geq 1}\frac{(-1)^{n+1}}{n} = \int_0^1\frac{\mathrm{d}x}{1+x} = \log 2,$$

we have:

$$\log\cos\frac{x}{2} = -\log 2+\sum_{n\geq 1}\frac{(-1)^{n+1}}{n}\cos(nx)\qquad \forall x\in(-\pi,\pi)$$

and by translating the variable:

$$\log\sin\frac{x}{2} = -\log 2-\sum_{n\geq 1}\frac{1}{n}\cos(nx)\qquad \forall x\in(0,2\pi)$$

$$\log\sin x = -\log 2 - \sum_{n\geq 1}\frac{1}{n}\cos(2nx) \qquad \forall x\in(0,\pi)$$

as wanted.
In fact proving from the right hand side is much clearer

$$\begin{aligned}
-\log 2 - \sum_{n=1}^{\infty}\frac{\cos 2nx}{n} &= -\log 2 - \Im\sum_{n=1}^{\infty}\frac{\mathrm{e}^{2\mathrm{i}nx}}{n}\\
&= -\log 2 - \Im\mathrm{Li}_1(\mathrm{e}^{2\mathrm{i}x})\\
&= -\log 2 + \Im\log(1-\mathrm{e}^{2\mathrm{i}x})\\
&= \log\sin x
\end{aligned}$$

From the series we have

$$\int_0^{\frac{\pi}{2}}\log\sin x\mathrm{d}x = \int_0^{\frac{\pi}{2}}\log\cos x\mathrm{d}x = -\frac{\pi}{2}\log 2$$

Similarly we have the Catalan Constant

$$\mathrm{G} = \int_0^{\frac{\pi}{4}}\log\cot x\mathrm{d}x = \beta(2) = 0.915965594\cdots$$

1.2 Properties

A

It's important to be acquainted with these specific formulas

$$\begin{aligned}
&\mathrm{Li}_0(x) = \frac{x}{1-x}\\
&\mathrm{Li}_1(x) = -\log(1-x)\\
&\mathrm{Li}_2(x) + \mathrm{Li}_2(\frac{x}{x-1}) = -\frac{1}{2}\log^2(1-x)\\
&\mathrm{Li}_2(x) + \mathrm{Li}_2(-x) = \frac{1}{2}\mathrm{Li}_2(x^2)\\
&\mathrm{Li}_2(x) + \mathrm{Li}_2(1-x) = \frac{\pi^2}{6} - \log x\log(1-x)
\end{aligned}$$

Take the third for example. To prove it, we simply need to differentiate on both sides:

$$\begin{aligned}
-\frac{\log(1-x)}{x} + \frac{\log(1-x)}{x} + \frac{\log(1-x)}{1-x} &= \frac{\log(1-x)}{1-x}\\
\mathrm{Li}_2(x) + \mathrm{Li}_2(\frac{x}{x-1}) &= -\frac{1}{2}\log^2(1-x) + C
\end{aligned}$$

Plug $x=0$ we get the demanded form. The other identities can be proved using the same method.

B

$$\mathrm{Li}_n(x)=\int_0^x\frac{\mathrm{Li}_{n-1}(x)}{x}\mathrm{d}x$$
$$\mathrm{Li}_n(x)=\frac{(-1)^{n-1}}{(n-1)!}\int_0^1\frac{x\log^{n-1}y}{1-xy}\mathrm{d}y$$

Let's prove them quickly:

$$\begin{aligned}\int_0^x\frac{\mathrm{Li}_{n-1}(x)}{x}\mathrm{d}x&=\int_0^x\frac{\sum_{k=1}^\infty\frac{x^k}{k^{n-1}}}{x}\mathrm{d}x\\&=\sum_{k=1}^\infty\frac{1}{k^{n-1}}\int_0^x x^{k-1}\mathrm{d}x\\&=\sum_{k=1}^\infty\frac{x^k}{k^n}\\&=\mathrm{Li}_n(x)\end{aligned}$$

$$\begin{aligned}\frac{(-1)^{n-1}}{(n-1)!}\int_0^1\frac{x\log^{n-1}y}{1-xy}\mathrm{d}y&=\frac{(-1)^{n-1}}{(n-1)!}\int_0^1 x\log^{n-1}y\sum_{k=0}^\infty(xy)^k\mathrm{d}y\\&=\frac{(-1)^{n-1}}{(n-1)!}\sum_{k=0}^\infty x^{k+1}\underbrace{\int_0^1 y^k\log^{n-1}y\mathrm{d}y}_{y\to\mathrm{e}^{-\frac{y}{k+1}}}\\&=\frac{1}{(n-1)!}\sum_{k=0}^\infty\frac{x^{k+1}}{(k+1)^n}\int_0^\infty\mathrm{e}^{-y}y^{n-1}\mathrm{d}y\\&=\frac{1}{(n-1)!}\sum_{k=1}^\infty\frac{x^k}{k^n}\Gamma(n)\\&=\mathrm{Li}_n(x)\end{aligned}$$

1.3 Practices

A001

$$\mathrm{Li}_2(\mathrm{i})=-\frac{\pi^2}{48}+\mathrm{iG}$$

Proof

$$\begin{aligned}\mathrm{Li}_2(\mathrm{i})&=\sum_{n=1}^\infty\frac{\mathrm{i}^n}{n^2}\\&=\sum_{n=1}^\infty\frac{(-1)^n}{(2n)^2}+\sum_{n=1}^\infty\frac{(-1)^{n-1}\mathrm{i}}{(2n-1)^2}\\&=-\frac{\pi^2}{48}+\mathrm{iG}\end{aligned}$$

A002

$$\mathrm{Li}_2(\frac{1}{1+\mathrm{i}}) = \frac{5\pi^2}{96} - \frac{\log^2 2}{8} + \mathrm{i}(\frac{\pi}{8}\log 2 - \mathrm{G})$$

Proof
Apply

$$\mathrm{Li}_2(x) + \mathrm{Li}_2(\frac{x}{x-1}) = -\frac{1}{2}\log^2(1-x)$$

We have

$$\mathrm{Li}_2(\frac{1-\mathrm{i}}{2}) + \mathrm{Li}_2(\mathrm{i}) = -\frac{1}{2}\log^2(\frac{1+\mathrm{i}}{2})$$
$$\mathrm{Li}_2(\frac{1-\mathrm{i}}{2}) - \frac{\pi^2}{48} + \mathrm{iG} = -\frac{1}{8}\log^2 2 + \frac{\pi^3}{32} + \mathrm{i}\frac{\pi\log 2}{8}$$
$$\mathrm{Li}_2(\frac{1}{1+\mathrm{i}}) = \frac{5\pi^2}{96} - \frac{\log^2 2}{8} + \mathrm{i}(\frac{\pi}{8}\log 2 - \mathrm{G})$$

A003

$$\mathrm{Li}_2(\frac{\sqrt{5}-1}{2}) = \frac{\pi^2}{10} - \log^2(\frac{1+\sqrt{5}}{2})$$

Proof
Apply

$$\mathrm{Li}_2(x) + \mathrm{Li}_2(\frac{x}{x-1}) = -\frac{1}{2}\log^2(1-x)$$
$$\mathrm{Li}_2(x) + \mathrm{Li}_2(-x) = \frac{1}{2}\mathrm{Li}_2(x^2)$$
$$\mathrm{Li}_2(x) + \mathrm{Li}_2(1-x) = \frac{\pi^2}{6} - \log x\log(1-x)$$

We have

$$\mathrm{Li}_2(\frac{3-\sqrt{5}}{2}) + \mathrm{Li}_2(\frac{1-\sqrt{5}}{2}) = -\frac{1}{2}\log^2(\frac{1+\sqrt{5}}{2})$$
$$\mathrm{Li}_2(\frac{\sqrt{5}-1}{2}) + \mathrm{Li}_2(\frac{1-\sqrt{5}}{2}) = \frac{1}{2}\mathrm{Li}_2(\frac{3-\sqrt{5}}{2})$$
$$\mathrm{Li}_2(\frac{\sqrt{5}-1}{2}) + \mathrm{Li}_2(\frac{3-\sqrt{5}}{2}) = \frac{\pi^2}{6} - 2\log^2(\frac{1+\sqrt{5}}{2})$$

Solve this system of equations

$$\mathrm{Li}_2(\frac{\sqrt{5}-1}{2}) = \frac{\pi^2}{10} - \log^2(\frac{1+\sqrt{5}}{2})$$

A004

$$\int_0^1 \frac{\log(1+x^2)}{x}\mathrm{d}x = \frac{\pi^2}{12}$$

Proof

$$\begin{aligned}\int_0^1 \frac{\log(1+x^2)}{x}\mathrm{d}x \xlongequal{x^2\to x} & \frac{1}{2}\int_0^1 \frac{\log(1+x)}{x}\mathrm{d}x \\ =& -\frac{1}{2}\mathrm{Li}_2(-1) \\ =& \frac{\pi^2}{12}\end{aligned}$$

A005

$$\int_0^1 \frac{\log^2(1+x^2)}{x^3}\mathrm{d}x = \frac{\pi^2}{12} - \log^2 2$$

Proof

$$\begin{aligned}\int_0^1 \frac{\log^2(1+x^2)}{x^3}\mathrm{d}x &= \int_0^1 \log^2(1+x^2)\mathrm{d}(-\frac{1}{2x^2}) \\ &= -\frac{1}{2}\log^2 2 + 2\int_0^1 (\frac{\log(1+x^2)}{x} - \frac{x\log(1+x^2)}{1+x^2})\mathrm{d}x \\ &= \frac{\pi^2}{12} - \log^2 2\end{aligned}$$

A006

$$\int_0^{\frac{\pi}{4}} \log\cos x\mathrm{d}x = \frac{\mathrm{G}}{2} - \frac{\pi}{4}\log 2$$
$$\int_0^{\frac{\pi}{4}} \log\sin x\mathrm{d}x = -\frac{\mathrm{G}}{2} - \frac{\pi}{4}\log 2$$

Proof
Set

$$I_1 = \int_0^{\frac{\pi}{4}} \log\cos x\mathrm{d}x$$
$$I_2 = \int_0^{\frac{\pi}{4}} \log\sin x\mathrm{d}x$$

Then

$$\begin{aligned}I_1 + I_2 &= \int_0^{\frac{\pi}{4}} \log(\frac{1}{2}\sin 2x)\mathrm{d}x \\ &= -\frac{\pi}{4}\log 2 + \frac{1}{2}\int_0^{\frac{\pi}{2}} \log\sin x\mathrm{d}x \\ &= -\frac{\pi}{2}\log 2\end{aligned}$$

$$\begin{aligned}I_1 - I_2 &= \int_0^{\frac{\pi}{4}} \log\cot x\mathrm{d}x \\ &= \mathrm{G}\end{aligned}$$

Solve this system of equations

$$\int_0^{\frac{\pi}{4}} \log\cos x\mathrm{d}x = \frac{\mathrm{G}}{2} - \frac{\pi}{4}\log 2$$
$$\int_0^{\frac{\pi}{4}} \log\sin x\mathrm{d}x = -\frac{\mathrm{G}}{2} - \frac{\pi}{4}\log 2$$

A007

$$\int_0^{\frac{\pi}{4}} x\log\cos x\mathrm{d}x = \frac{\pi\mathrm{G}}{8} - \frac{\pi^2}{32}\log 2 - \frac{21}{128}\zeta(3)$$

Proof

$$\begin{aligned}
\int_0^{\frac{\pi}{4}} x\log\cos x\mathrm{d}x &= -\int_0^{\frac{\pi}{4}} x(\log 2 + \sum_{n=1}^{\infty}\frac{(-1)^n\cos 2nx}{n})\mathrm{d}x \\
&= -\log 2\int_0^{\frac{\pi}{4}} x\mathrm{d}x - \sum_{n=1}^{\infty}\frac{(-1)^n}{n}\int_0^{\frac{\pi}{4}} x\cos 2nx\mathrm{d}x \\
&= -\frac{\pi^2}{32}\log 2 - \sum_{n=1}^{\infty}\frac{(-1)^n}{n}(\frac{\pi\sin(\frac{n\pi}{2})}{8n} + \frac{\cos(\frac{n\pi}{2})}{4n^2} - \frac{1}{4n^2}) \\
&= -\frac{\pi^2}{32}\log 2 - \frac{\pi}{8}\sum_{n=1}^{\infty}\frac{(-1)^n\sin(\frac{n\pi}{2})}{n^2} - \frac{1}{4}\sum_{n=1}^{\infty}\frac{(-1)^n\cos(\frac{n\pi}{2})}{n^3} + \frac{1}{4}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^3} \\
&= -\frac{\pi^2}{32}\log 2 - \frac{\pi}{8}\Im\mathrm{Li}_2(-\mathrm{i}) - \frac{1}{4}\Re\mathrm{Li}_3(-\mathrm{i}) + \frac{1}{4}\mathrm{Li}_3(-1) \\
&= \frac{\pi\mathrm{G}}{8} - \frac{\pi^2}{32}\log 2 - \frac{21}{128}\zeta(3)
\end{aligned}$$

A008

$$\int_0^{\frac{\pi}{4}} x\log(1+\tan x)\mathrm{d}x = \frac{\pi^2}{64}\log 2 - \frac{\pi\mathrm{G}}{8} + \frac{21\zeta(3)}{64}$$

Proof

$$\begin{aligned}\int_0^{\frac{\pi}{4}} x\log(1+\tan x)\mathrm{d}x &= \int_0^{\frac{\pi}{4}} x\log(\sqrt{2}\sin(x+\frac{\pi}{4}))\mathrm{d}x - \int_0^{\frac{\pi}{4}} x\log\cos x\mathrm{d}x \\ &= \frac{\pi^2}{64}\log 2 + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (x-\frac{\pi}{4})\log\sin x\mathrm{d}x - \int_0^{\frac{\pi}{4}} x\log\cos x\mathrm{d}x \\ &= \frac{\pi^2}{64}\log 2 + \int_0^{\frac{\pi}{4}} (\frac{\pi}{4}-x)\log\cos x\mathrm{d}x - \int_0^{\frac{\pi}{4}} x\log\cos x\mathrm{d}x \\ &= \frac{\pi^2}{64}\log 2 + \frac{\pi}{4}\int_0^{\frac{\pi}{4}} \log\cos x\mathrm{d}x - 2\int_0^{\frac{\pi}{4}} x\log\cos x\mathrm{d}x \\ &= \frac{\pi^2}{64}\log 2 - \frac{\pi\mathrm{G}}{8} + \frac{21\zeta(3)}{64}\end{aligned}$$

A009

$$\int_0^{\frac{\pi}{2}} x^2\cot x\mathrm{d}x = \frac{1}{4}\pi^2\log 2 - \frac{7}{8}\zeta(3)$$

Proof
Consider

$$\cot x = -\frac{\mathrm{i}(\mathrm{e}^{-\mathrm{i}x}+\mathrm{e}^{\mathrm{i}x})}{\mathrm{e}^{-\mathrm{i}x}-\mathrm{e}^{\mathrm{i}x}} = -\mathrm{i} - 2\mathrm{i}\mathrm{Li}_0(\mathrm{e}^{2\mathrm{i}x})$$

We have

$$\begin{aligned}\int x^2\cot x\mathrm{d}x &= -\mathrm{i}\int x^2\mathrm{d}x - 2\mathrm{i}\int x^2\mathrm{Li}_0(\mathrm{e}^{2\mathrm{i}x})\mathrm{d}x \\ &= -\frac{\mathrm{i}x^3}{3} - \int x^2\mathrm{d}(\mathrm{Li}_1(\mathrm{e}^{2\mathrm{i}x})) \\ &= -\frac{\mathrm{i}x^3}{3} - x^2\mathrm{Li}_1(\mathrm{e}^{2\mathrm{i}x}) + 2\int x\mathrm{Li}_1(\mathrm{e}^{2\mathrm{i}x})\mathrm{d}x \\ &= -\frac{\mathrm{i}x^3}{3} - x^2\mathrm{Li}_1(\mathrm{e}^{2\mathrm{i}x}) - \mathrm{i}x\mathrm{Li}_2(\mathrm{e}^{2\mathrm{i}x}) + \mathrm{i}\int \mathrm{Li}_2(\mathrm{e}^{2\mathrm{i}x})\mathrm{d}x \\ &= -\frac{\mathrm{i}x^3}{3} - x^2\mathrm{Li}_1(\mathrm{e}^{2\mathrm{i}x}) - \mathrm{i}x\mathrm{Li}_2(\mathrm{e}^{2\mathrm{i}x}) + \frac{1}{2}\mathrm{Li}_3(\mathrm{e}^{2\mathrm{i}x}) + C\end{aligned}$$

Hence

$$\int_0^{\frac{\pi}{2}} x^2\cot x\mathrm{d}x = \frac{1}{4}\pi^2\log 2 - \frac{7}{8}\zeta(3)$$

A010

$$\int_0^{\frac{\pi}{4}} x^2\tan^2 x\mathrm{d}x = -\mathrm{G} + \frac{\pi^2}{16} - \frac{\pi^3}{192} + \frac{\pi}{4}\log 2$$

Proof
Consider

$$\begin{aligned}(x^2\tan x)' &= 2x\tan x + x^2\sec^2 x\\ &= x^2 + 2x\tan x + x^2\tan^2 x\\ \int x^2\tan^2 x\mathrm{d}x &= x^2\tan x - \int x^2\mathrm{d}x - 2\int x\tan x\mathrm{d}x\end{aligned}$$

We have

$$\begin{aligned}\int_0^{\frac{\pi}{4}} x^2\tan^2 x\mathrm{d}x &= x^2\tan x - \frac{x^3}{3} - 2\mathrm{i}\int x(1+2\mathrm{Li}_0(-\mathrm{e}^{2\mathrm{i}x}))\mathrm{d}x\\ &= x^2\tan x - \frac{x^3}{3} - \mathrm{i}x^2 - 2\int x\mathrm{d}(\mathrm{Li}_1(-\mathrm{e}^{2\mathrm{i}x}))\\ &= x^2\tan x - \frac{x^3}{3} - \mathrm{i}x^2 - 2x\mathrm{Li}_1(-\mathrm{e}^{2\mathrm{i}x}) - \mathrm{i}\mathrm{Li}_2(-\mathrm{e}^{2\mathrm{i}x}) + C\end{aligned}$$

Hence

$$\int_0^{\frac{\pi}{4}} x^2\tan^2 x\mathrm{d}x = -\mathrm{G} + \frac{\pi^2}{16} - \frac{\pi^3}{192} + \frac{\pi}{4}\log 2$$

A011

$$\int_0^1 \frac{\mathrm{Li}_2(x)}{x\sqrt{1-x^2}}\mathrm{d}x = \frac{3}{8}\pi^2\log 2 - \frac{7}{16}\zeta(3)$$

Proof

$$\begin{aligned}\int_0^1 \frac{\mathrm{Li}_2(x)}{x\sqrt{1-x^2}}\mathrm{d}x &= -\int_0^1\int_0^1 \frac{1}{x\sqrt{1-x^2}}\frac{x\log y}{1-xy}\mathrm{d}y\mathrm{d}x\\ &= -\int_0^1 \log y\int_0^1 \frac{1}{(1-xy)\sqrt{1-x^2}}\mathrm{d}x\mathrm{d}y\\ &= -\int_0^1 \frac{\pi-\arccos y}{\sqrt{1-y^2}}\log y\mathrm{d}y\\ &= -\int_0^1 \frac{\pi-\frac{\pi}{2}+\arcsin x}{\sqrt{1-y^2}}\log y\mathrm{d}y\\ &= -\frac{\pi}{2}\int_0^1 \frac{\log y}{\sqrt{1-y^2}}\mathrm{d}y - \int_0^1 \frac{\arcsin x}{\sqrt{1-y^2}}\log y\mathrm{d}y\\ &= \frac{1}{4}\pi^2\log 2 + \frac{1}{2}\int_0^1 \frac{\arcsin^2 y}{y}\mathrm{d}y\\ &= \frac{1}{4}\pi^2\log 2 + \frac{1}{2}\int_0^{\frac{\pi}{2}} y^2\cot y\mathrm{d}y\\ &= \frac{3}{8}\pi^2\log 2 - \frac{7}{16}\zeta(3)\end{aligned}$$

A012

$$\int_0^1 \frac{\arctan x}{x+1}\log x\mathrm{d}x = -\frac{\pi^3}{64}+\frac{\mathrm{G}}{2}\log 2$$

Proof

$$\begin{aligned}
\mathrm{LHS} &= \int_0^1 \arctan x \mathrm{d}(\mathrm{Li}_2(-x)+\log x\log(x+1)) \\
&= -\frac{\pi^3}{48}-\int_0^1 \frac{\log x\log(x+1)}{x^2+1}\mathrm{d}x-\int_0^1\frac{\mathrm{Li}_2(-x)}{x^2+1}\mathrm{d}x \\
&= -\frac{\pi^3}{48}-\int_0^1 \frac{\log x\log(x+1)}{x^2+1}\mathrm{d}x-\int_0^1\frac{1}{x^2+1}\int_0^1\frac{x\log y}{xy+1}\mathrm{d}y\mathrm{d}x \\
&= -\frac{\pi^3}{48}-\int_0^1 \frac{\log x\log(x+1)}{x^2+1}\mathrm{d}x-\int_0^1\log y\int_0^1\frac{x}{(x^2+1)(xy+1)}\mathrm{d}x\mathrm{d}y \\
&= -\frac{\pi^3}{48}-\int_0^1 \frac{\log x\log(x+1)}{x^2+1}\mathrm{d}x-\int_0^1\log y(\frac{\pi y}{4(y^2+1)}+\frac{\log 2}{2(y^2+1)}-\frac{\log(y+1)}{y^2+1})\mathrm{d}y \\
&= -\frac{\pi^3}{48}-\frac{\pi}{4}\underbrace{\int_0^1\frac{y\log y}{y^2+1}\mathrm{d}y}_{y\to \mathrm{e}^{-y}}-\frac{\log 2}{2}\underbrace{\int_0^1\frac{\log y}{y^2+1}\mathrm{d}y}_{y\to \mathrm{e}^{-y}} \\
&= -\frac{\pi^3}{48}+\frac{\pi}{4}\int_0^\infty\frac{y}{\mathrm{e}^{2y}+1}\mathrm{d}y+\frac{\log 2}{2}\int_0^\infty\frac{x}{\mathrm{e}^x+\mathrm{e}^{-x}}\mathrm{d}x \\
&= -\frac{\pi^3}{48}+\frac{\pi}{16}\eta(2)\Gamma(2)+\frac{\log 2}{2}\beta(2)\Gamma(2) \\
&= -\frac{\pi^3}{64}+\frac{\mathrm{G}}{2}\log 2 \\
&= \mathrm{RHS}
\end{aligned}$$

A013

$$\int_0^1 \frac{\arctan x}{x+1}\log^2 x\mathrm{d}x = -\frac{\pi^2}{24}\mathrm{G}-\frac{\pi^3}{32}\log 2+\frac{21\pi}{64}\zeta(3)$$

Proof

$$\begin{aligned}
\mathrm{LHS} &= \int_0^1 \arctan x \mathrm{d}(-2\mathrm{Li}_3(-x)+2\log x\mathrm{Li}_2(-x)+\log(x+1)\log^2 x) \\
&= \frac{3\pi}{8}\zeta(3)+\int_0^1\frac{2\mathrm{Li}_3(-x)-2\log x\mathrm{Li}_2(-x)-\log(x+1)\log^2 x}{x^2+1}\mathrm{d}x \\
&= \frac{3\pi}{8}\zeta(3)-\underbrace{\int_0^1\int_0^1\frac{x\log^2 y}{(xy+1)(x^2+1)}\mathrm{d}y\mathrm{d}x}_{\mathcal{J}}-2\underbrace{\int_0^1\int_0^1\frac{x\log x\log y}{(xy+1)(x^2+1)}\mathrm{d}y\mathrm{d}x}_{\mathcal{K}}-\underbrace{\int_0^1\frac{\log(x+1)\log^2 x}{x^2+1}\mathrm{d}x}_{\mathcal{M}}
\end{aligned}$$

$$
\begin{aligned}
\mathcal{J} &= \int_0^1 \log^2 y(\frac{\pi y}{4(y^2+1)} + \frac{\log 2}{2(y^2+1)} - \frac{\log(y+1)}{y^2+1})\mathrm{d}y \\
&= \frac{\pi}{4}\underbrace{\int_0^1 \frac{y\log^2 y}{y^2+1}\mathrm{d}y}_{y\to \mathrm{e}^{-y}} + \frac{\log 2}{2}\underbrace{\int_0^1 \frac{\log^2 y}{y^2+1}\mathrm{d}y}_{y\to \mathrm{e}^{-y}} - \int_0^1 \frac{\log^2 y\log(y+1)}{y^2+1})\mathrm{d}y \\
&= \frac{\pi}{4}\int_0^\infty \frac{y^2}{\mathrm{e}^{2y}+1}\mathrm{d}y + \frac{\log 2}{2}\int_0^\infty \frac{y^2}{\mathrm{e}^y+\mathrm{e}^{-y}}\mathrm{d}y - \mathcal{M} \\
&= \frac{\pi}{32}\eta(3)\Gamma(3) + \frac{\log 2}{2}\beta(3)\Gamma(3) - \mathcal{M} \\
&= \frac{3\pi}{64}\zeta(3) + \frac{\pi^3}{32}\log 2 - \mathcal{M} \\
\mathcal{K} &= \int_0^1 \log y\int_0^1 \log x(\frac{x+y}{(x^2+1)(y^2+1)} - \frac{y}{(xy+1)(y^2+1)})\mathrm{d}x\mathrm{d}y \\
&= \int_0^1 \log y(\frac{1}{y^2+1}\int_0^1 \frac{x\log x}{x^2+1}\mathrm{d}x + \frac{y}{y^2+1}\int_0^1 \frac{\log x}{x^2+1}\mathrm{d}x - \frac{y}{y^2+1}\underbrace{\int_0^1 \frac{\log x}{xy+1}\mathrm{d}x}_{\mathcal{N}})\mathrm{d}y \\
\mathcal{N} &= \int_0^1 \log x\sum_{n=0}^\infty (-1)^n(xy)^n\mathrm{d}x \\
&= \sum_{n=0}^\infty (-y)^n\int_0^1 x^n\log x\mathrm{d}x \\
&= -\sum_{n=0}^\infty \frac{(-y)^n}{(n+1)^2} \\
&= \frac{1}{y}\mathrm{Li}_2(-y) \\
\mathcal{K} &= -\frac{\pi^2}{48}\int_0^1 \frac{\log y}{y^2+1}\mathrm{d}y - \mathrm{G}\int_0^1 \frac{y\log y}{y^2+1}\mathrm{d}y - \int_0^1 \frac{\log y\mathrm{Li}_2(-y)}{y^2+1}\mathrm{d}y \\
&= \frac{\pi^2}{24}\mathrm{G} - \int_0^1 \frac{\log y}{y^2+1}\int_0^1 \frac{y\log x}{xy+1}\mathrm{d}x\mathrm{d}y \\
&= \frac{\pi^2}{24}\mathrm{G} - \mathcal{K} \\
\mathcal{K} &= \frac{\pi^2}{48}\mathrm{G} \\
\mathrm{LHS} &= \frac{3\pi}{8}\zeta(3) - \mathcal{J} - 2\mathcal{K} - \mathcal{M} \\
&= -\frac{\pi^2}{24}\mathrm{G} - \frac{\pi^3}{32}\log 2 + \frac{21\pi}{64}\zeta(3) \\
&= \mathrm{RHS}
\end{aligned}
$$

A014

$$\int_0^1 \frac{\log x \log(x+1)}{x^2+1}\mathrm{d}x = 2\Im(\mathrm{Li}_3(1+\mathrm{i})) - \Im(\mathrm{Li}_3(\frac{1+\mathrm{i}}{2})) - 2\mathrm{G}\log 2 - \frac{3\pi\log^2 2}{32} - \frac{3\pi^3}{128}$$

Proof

$$\mathrm{LHS} = -\frac{1}{2}\underbrace{\int_0^1 \frac{\log^2(\frac{x}{x+1})}{x^2+1}\mathrm{d}x}_{\mathcal{J}} + \frac{1}{2}\underbrace{\int_0^1 \frac{\log^2 x}{x^2+1}\mathrm{d}x}_{\mathcal{K}} + \frac{1}{2}\underbrace{\int_0^1 \frac{\log^2(x+1)}{x^2+1}\mathrm{d}x}_{\mathcal{M}}$$

$$\begin{aligned}
\mathcal{J} \overset{\frac{x}{x+1}\to x}{=\!=\!=} & \int_0^{\frac{1}{2}} \frac{\log^2 x}{2x^2-2x+1}\mathrm{d}x \\
=& \Im\int_0^{\frac{1}{2}} \frac{\log^2 x}{x-\frac{1+\mathrm{i}}{2}}\mathrm{d}x \\
=& -\Im\int_0^{\frac{1}{2}} \frac{2\log^2 x}{1+\mathrm{i}}\sum_{n=0}^{\infty}(\frac{2x}{1+\mathrm{i}})^n\mathrm{d}x \\
=& -\Im\sum_{n=0}^{\infty}(\frac{2}{1+\mathrm{i}})^{n+1}\int_0^{\frac{1}{2}} x^n\log^2 x\mathrm{d}x \\
=& -\Im\sum_{n=0}^{\infty}(\frac{2}{1+\mathrm{i}})^{n+1}(\frac{x^{n+1}\log^2 x}{n+1} - \frac{2x^{n+1}\log x}{(n+1)^2} + \frac{2x^{n+1}}{(n+1)^3})\Big|_0^{\frac{1}{2}} \\
=& -\Im(\log^2 2\sum_{n=0}^{\infty}\frac{(\frac{1}{1+\mathrm{i}})^{n+1}}{n+1} + 2\log 2\sum_{n=0}^{\infty}\frac{(\frac{1}{1+\mathrm{i}})^{n+1}}{(n+1)^2} + 2\sum_{n=0}^{\infty}\frac{(\frac{1}{1+\mathrm{i}})^{n+1}}{(n+1)^3}) \\
=& -\Im(\log^2 2\mathrm{Li}_1(\frac{1}{1+\mathrm{i}}) + 2\log 2\mathrm{Li}_2(\frac{1}{1+\mathrm{i}}) + 2\mathrm{Li}_3(\frac{1}{1+\mathrm{i}})) \\
=& 2\mathrm{G}\log 2 + 2\Im(\mathrm{Li}_3(\frac{1+\mathrm{i}}{2}))
\end{aligned}$$

$$\begin{aligned}
\mathcal{K} &= \Im\int_0^1 \frac{\log^2 x}{x-\mathrm{i}}\mathrm{d}x \\
&= -\Im(-\mathrm{i}\int_0^1 \frac{\log^2 x}{1-\frac{x}{\mathrm{i}}}\mathrm{d}x) \\
&= -\Im(\sum_{n=0}^{\infty}(-\mathrm{i})^{n+1}\int_0^1 x^n\log^2 x\mathrm{d}x) \\
&= -\Im(2\mathrm{Li}_3(-\mathrm{i})) \\
&= \frac{\pi^3}{16}
\end{aligned}$$

$$\begin{aligned}
\mathcal{M} &= \Im \int_0^1 \frac{\log^2(1+x)}{x-\mathrm{i}}\mathrm{d}x \\
&\overset{x+1\to x}{=\!=\!=} \Im \int_1^2 \frac{\log^2 x}{x-\mathrm{i}}\mathrm{d}x \\
&= -\Im(\frac{1}{1+\mathrm{i}}\int_1^2 \frac{\log^2 x}{1-\frac{x}{1+\mathrm{i}}}\mathrm{d}x) \\
&= -\Im(\sum_{n=0}^{\infty}(\frac{1}{1+\mathrm{i}})^{n+1}\int_1^2 x^n\log^2 x\mathrm{d}x) \\
&= 4\Im(\mathrm{Li}_3(1+\mathrm{i})) - \frac{7\pi^3}{64} - \frac{3\pi\log^2 2}{16} - 2\mathrm{G}\log 2 \\
\mathrm{LHS} &= -\frac{1}{2}\mathcal{J} + \frac{1}{2}\mathcal{K} + \frac{1}{2}\mathcal{M} \\
&= 2\Im(\mathrm{Li}_3(1+\mathrm{i})) - \Im(\mathrm{Li}_3(\frac{1+\mathrm{i}}{2})) - 2\mathrm{G}\log 2 - \frac{3\pi\log^2 2}{32} - \frac{3\pi^3}{128} \\
&= \mathrm{RHS}
\end{aligned}$$

A015

$$\int_0^{\frac{\pi}{2}} x\log^2\sin x\mathrm{d}x = \mathrm{Li}_4(\frac{1}{2}) - \frac{19\pi^4}{2880} + \frac{1}{24}\log^4 2 + \frac{1}{12}\pi^2\log^2 2$$

Proof
Introduce a dual integral for its simpler form:

$$\int_0^{\frac{\pi}{2}} x\log^2\cos x\mathrm{d}x$$

Note that

$$\log(\frac{1+\mathrm{e}^{2\mathrm{i}x}}{2}) = \log\cos x + \mathrm{i}x$$
$$\Im(\log^3(\frac{1+\mathrm{e}^{2\mathrm{i}x}}{2})) = 3x\log^2\cos x - x^3$$

We have

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} x\log^2\cos x\mathrm{d}x &= \frac{1}{3}\int_0^{\frac{\pi}{2}} x^3\mathrm{d}x + \frac{1}{3}\Im\int_0^{\frac{\pi}{2}}\log^3(\frac{1+\mathrm{e}^{2\mathrm{i}x}}{2})\mathrm{d}x \\
&= \frac{\pi^4}{192} + \frac{1}{6}\Im\int_0^{\pi}\log^3(\frac{1+\mathrm{e}^{\mathrm{i}x}}{2})\mathrm{d}x \\
&= \frac{\pi^4}{192} - \frac{1}{6}\Re\int_\Gamma \frac{\log^3(\frac{1+z}{2})}{z}\mathrm{d}z
\end{aligned}$$

Γ represents the upper half of the unit circle, apply the Residue formula:

$$\begin{aligned}\Re\int_\Gamma \frac{\log^3(\frac{1+z}{2})}{z}\mathrm{d}z =& \int_0^1 \frac{1}{x}(\log^3(\frac{1-x}{2})-\log^3(\frac{1+x}{2}))\mathrm{d}x\\ =&3\log^2 2\int_0^1\frac{\log(1-x)}{x}\mathrm{d}x-3\log 2\int_0^1\frac{\log^2(1-x)}{x}\mathrm{d}x+\int_0^1\frac{\log^3(1-x)}{x}\mathrm{d}x\\ &-3\log^2 2\int_0^1\frac{\log(1+x)}{x}\mathrm{d}x+3\log 2\int_0^1\frac{\log^2(1+x)}{x}\mathrm{d}x-\int_0^1\frac{\log^3(1+x)}{x}\mathrm{d}x\\ =&-\frac{2\pi^4}{15}-\pi^2\log^2 2+\frac{\log^4 2}{4}+6\mathrm{Li}_4(\frac{1}{2})\end{aligned}$$

Hence

$$\int_0^{\frac{\pi}{2}} x\log^2\cos x\mathrm{d}x=\frac{79\pi^4}{2880}+\frac{\pi^2\log^2 2}{6}-\frac{\log^4 2}{24}-\mathrm{Li}_4(\frac{1}{2})$$

Since

$$\int_0^{\frac{\pi}{2}} x\log^2\sin x\mathrm{d}x=\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\log^2\cos x\mathrm{d}x-\int_0^{\frac{\pi}{2}} x\log^2\cos x\mathrm{d}x$$

Where

$$\begin{aligned}\int_0^{\frac{\pi}{2}}\log^2\cos x\mathrm{d}x=&\int_0^{\frac{\pi}{2}}x^2\mathrm{d}x+\Re\int_0^{\frac{\pi}{2}}\log^2(\frac{1+\mathrm{e}^{2\mathrm{i}x}}{2})\mathrm{d}x\\ =&\frac{\pi^3}{24}+\frac{1}{2}\Im\mathrm{i}\pi\log^2 2\\ =&\frac{\pi^3}{24}+\frac{1}{2}\pi\log^2 2\end{aligned}$$

We proved

$$\int_0^{\frac{\pi}{2}} x\log^2\sin x\mathrm{d}x=\mathrm{Li}_4(\frac{1}{2})-\frac{19\pi^4}{2880}+\frac{1}{24}\log^4 2+\frac{1}{12}\pi^2\log^2 2$$

A016

$$\int_0^1\frac{\mathrm{Li}_3(x)}{x\sqrt{1-x^2}}\mathrm{d}x=\frac{1}{2}\mathrm{Li}_4(\frac{1}{2})+\frac{41\pi^4}{5760}+\frac{1}{48}\log^4 2+\frac{1}{6}\pi^2\log^2 2$$

Proof

$$\begin{aligned}
\int_0^1 \frac{\mathrm{Li}_3(x)}{x\sqrt{1-x^2}}\mathrm{d}x &= \frac{1}{2}\int_0^1\int_0^1 \frac{1}{x\sqrt{1-x^2}}\frac{x\log^2 y}{1-xy}\mathrm{d}y\mathrm{d}x \\
&= \frac{1}{2}\int_0^1 \log^2 y \int_0^1 \frac{1}{(1-xy)\sqrt{1-x^2}}\mathrm{d}x\mathrm{d}y \\
&= \frac{1}{2}\int_0^1 \frac{\pi-\arccos y}{\sqrt{1-y^2}}\log^2 y\mathrm{d}y \\
&= \frac{1}{2}\int_0^1 \frac{\pi-\frac{\pi}{2}+\arcsin x}{\sqrt{1-y^2}}\log^2 y\mathrm{d}y \\
&= \frac{\pi}{4}\int_0^1 \frac{\log^2 y}{\sqrt{1-y^2}}\mathrm{d}y + \frac{1}{2}\int_0^1 \frac{\arcsin x}{\sqrt{1-y^2}}\log^2 y\mathrm{d}y \\
&= \frac{\pi^4}{96}+\frac{1}{8}\pi^2\log^2 2 - \frac{1}{2}\int_0^1 \frac{\arcsin^2 y\log y}{y}\mathrm{d}y \\
&= \frac{\pi^4}{96}+\frac{1}{8}\pi^2\log^2 2 - \frac{1}{2}\int_0^{\frac{\pi}{2}} y^2\log\sin y\cot y\mathrm{d}y \\
&= \frac{\pi^4}{96}+\frac{1}{8}\pi^2\log^2 2 + \frac{1}{2}\int_0^{\frac{\pi}{2}} y\log^2\sin y\mathrm{d}y \\
&= \frac{1}{2}\mathrm{Li}_4(\frac{1}{2})+\frac{41\pi^4}{5760}+\frac{1}{48}\log^4 2+\frac{1}{6}\pi^2\log^2 2
\end{aligned}$$

1.4 More Practices

B001

$$\int_0^1 \frac{\log(1-x)\log x}{1-x}\mathrm{d}x$$

$$\begin{aligned}
I &= \int_0^1 \log x \mathrm{d}(-\frac{1}{2}\log^2(1-x)) \\
&= \frac{1}{2}\int_0^1 \frac{\log^2(1-x)}{x}\mathrm{d}x \\
&= \frac{1}{2}\int_0^1 \frac{\log^2 x}{1-x}\mathrm{d}x \\
&= \frac{1}{2}\int_0^\infty \frac{x^2}{\mathrm{e}^x-1}\mathrm{d}x \\
&= \frac{1}{2}\Gamma(3)\zeta(3) \\
&= \zeta(3)
\end{aligned}$$

B002

$$\int_0^1 \frac{\log x \log(1+x)}{1-x}\mathrm{d}x$$

$$\begin{aligned}
I &= \int_0^1 \frac{\log x \log(1-x^2)}{1-x}\mathrm{d}x - \int_0^1 \frac{\log x \log(1-x)}{1-x}\mathrm{d}x \\
&= \underbrace{\int_0^1 \frac{x\log x \log(1-x^2)}{1-x^2}}_{x^2\to x} + \underbrace{\int_0^1 \frac{\log x \log(1-x)}{1-x^2}\mathrm{d}x}_{x^2\to x} - \zeta(3) \\
&= \frac{1}{4}\int_0^1 \frac{\log x \log(1-x)}{1-x}\mathrm{d}x + \frac{1}{4}\int_0^1 \frac{x^{-\frac{1}{2}}\log x \log(1-x)}{1-x}\mathrm{d}x - \zeta(3) \\
&= -\frac{3}{4}\zeta(3) + \frac{1}{4}\frac{\partial^2 \mathrm{B}(p,q)}{\partial p \partial q}\bigg|_{p=\frac{1}{2},q\to 0^+} \\
&= \zeta(3) - \frac{\pi^2}{4}\log 2
\end{aligned}$$

B003

$$\sum_{n=1}^\infty \frac{H_n}{n^3}$$

(Euler Sum)

$$
\begin{aligned}
S &= \sum_{n=1}^{\infty} \frac{1}{n^3} \int_0^1 \frac{1-t^n}{1-t} \mathrm{d}t \\
&= \int_0^1 (\frac{1}{1-t} \sum_{n=1}^{\infty} \frac{1-t^n}{n^3}) \mathrm{d}t \\
&= \int_0^1 \frac{\zeta(3) - \mathrm{Li}_3(t)}{1-t} \mathrm{d}t \\
&= -(\zeta(3) - \mathrm{Li}_3(t)) \log(1-t) \Big|_0^1 - \int_0^1 \frac{\mathrm{Li}_2(t) \log(1-t)}{t} \mathrm{d}t \\
&= \int_0^1 \mathrm{Li}_2(t) \mathrm{d}(\mathrm{Li}_2(t)) \\
&= \frac{\pi^4}{72}
\end{aligned}
$$

B004

$$
\sum_{n=1}^{\infty} \frac{H_n}{n^4 2^n}
$$

Let

$$
\mathcal{S} = \sum_{n=1}^{\infty} \frac{H_n}{n^4 2^n}
$$

We first consider a slightly different yet related sum

$$
\sum_{n=1}^{\infty} \frac{(-1)^n H_n}{n^4}
$$

Note that

$$
\int_0^1 x^{n-1} \log^3 x \mathrm{d}x = \frac{\partial^3}{\partial n^3} \int_0^1 x^{n-1} \mathrm{d}x = -\frac{6}{n^4}
$$

$$
\sum_{n=1}^{\infty} (-x)^{n-1} H_n = \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} (-x)^{n-k} \frac{(-x)^k}{k+1} = \frac{\log(1+x)}{x(1+x)}
$$

We have

$$
\begin{aligned}
\sum_{n=1}^{\infty} \frac{(-1)^n H_n}{n^4} &= \frac{1}{6} \int_0^1 \frac{\log^3 x \log(1+x)}{x(1+x)} \mathrm{d}x \\
&= \frac{1}{6} \int_0^1 \frac{\log^3 x \log(1+x)}{x} \mathrm{d}x - \frac{1}{6} \int_0^1 \frac{\log^3 x \log(1+x)}{1+x} \mathrm{d}x
\end{aligned}
$$

Therefore

$$
\frac{1}{6} \int_0^1 \frac{\log^3 x \log(1+x)}{x} \mathrm{d}x = \frac{1}{6} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \int_0^1 x^{n-1} \log^3 x \mathrm{d}x = -\eta(5)
$$

Apply a tricky transform:

$$\begin{aligned}\frac{1}{6}\int_0^1 \frac{\log^3 x \log(1+x)}{1+x}\mathrm{d}x &= \frac{1}{6}\underbrace{\int_1^2 \frac{\log^3(x-1)\log x}{x}\mathrm{d}x}_{x-1\to\frac{1-x}{x}} \\ &= \frac{1}{6}\int_{\frac{1}{2}}^1 \frac{\log x \log^3(\frac{x}{1-x})}{x}\mathrm{d}x \\ &= \frac{1}{6}\int_{\frac{1}{2}}^1 \frac{\log x(\log x - \log(1-x))^3}{x}\mathrm{d}x\end{aligned}$$

Simplify after expanding:

$$\begin{aligned}\sum_{n=1}^{\infty}\frac{(-1)^n H_n}{n^4} = -\eta(5) &+ \underbrace{\int_{\frac{1}{2}}^1 \frac{\log x \log^3(1-x)}{6x}\mathrm{d}x}_{\mathcal{I}_1} - \underbrace{\int_{\frac{1}{2}}^1 \frac{\log^2 x \log^2(1-x)}{2x}\mathrm{d}x}_{\mathcal{I}_2} \\ &+ \underbrace{\int_{\frac{1}{2}}^1 \frac{\log^3 x \log(1-x)}{2x}\mathrm{d}x}_{\mathcal{I}_3} - \underbrace{\int_{\frac{1}{2}}^1 \frac{\log^4 x}{6x}\mathrm{d}x}_{\mathcal{I}_4}\end{aligned}$$

Starting with the easiest integral,

$$\mathcal{I}_4 = \frac{1}{30}\log^5 2$$

For $\mathcal{I}_3$,

$$\begin{aligned}\mathcal{I}_3 =& -\frac{1}{2}\sum_{n=1}^{\infty}\frac{1}{n}\int_{\frac{1}{2}}^1 x^{n-1}\log^3 x\,\mathrm{d}x \\ =& -\frac{1}{2}\sum_{n=1}^{\infty}\frac{1}{n}\frac{\partial^3}{\partial n^3}\left(\frac{1}{n}-\frac{1}{n2^n}\right) \\ =& \sum_{n=1}^{\infty}\left(\frac{3}{n^5}-\frac{3}{n^5 2^n}-\frac{3\log 2}{n^4 2^n}-\frac{3\log^2 2}{n^3 2^{n+1}}-\frac{\log^3 2}{n^2 2^{n+1}}\right) \\ =& 3\zeta(5)-3\mathrm{Li}_5\left(\frac{1}{2}\right)-3\mathrm{Li}_4\left(\frac{1}{2}\right)\log 2-\frac{3}{2}\log^2 2\left(\frac{7}{8}\zeta(3)-\frac{\pi^2}{12}\log 2+\frac{1}{6}\log^3 2\right) \\ &-\frac{1}{2}\log^3 2\left(\frac{\pi^2}{12}-\frac{1}{2}\log^2 2\right) \\ =& 3\zeta(5)-3\mathrm{Li}_5\left(\frac{1}{2}\right)-3\mathrm{Li}_4\left(\frac{1}{2}\right)\log 2-\frac{21}{16}\zeta(3)\log^2 2+\frac{\pi^2}{12}\log^3 2\end{aligned}$$

For $\mathcal{I}_2$,

$$\begin{aligned}
\mathcal{I}_2 =&\frac{1}{6}\log^5 2+\frac{1}{3}\int_{\frac{1}{2}}^{1}\frac{\log^3 x\log(1-x)}{1-x}\mathrm{d}x=\frac{1}{6}\log^5 2-\frac{1}{3}\sum_{n=1}^{\infty}H_n\frac{\partial^3}{\partial n^3}\left(\frac{1}{n+1}-\frac{1}{(n+1)2^{n+1}}\right)\\
=&\frac{1}{6}\log^5 2+\sum_{n=1}^{\infty}\frac{2H_n}{(n+1)^4}-\sum_{n=1}^{\infty}\frac{2H_n}{(n+1)^4 2^{n+1}}-\sum_{n=1}^{\infty}\frac{2\log 2H_n}{(n+1)^3 2^{n+1}}\\
&-\sum_{n=1}^{\infty}\frac{\log^2 2H_n}{(n+1)^2 2^{n+1}}-\sum_{n=1}^{\infty}\frac{\log^3 2H_n}{3(n+1)2^{n+1}}\\
=&\frac{1}{6}\log^5 2+4\zeta(5)-\frac{\pi^2}{3}\zeta(3)-2\mathcal{S}+2\mathrm{Li}_5\left(\frac{1}{2}\right)-\frac{\pi^4}{360}\log 2+\frac{1}{4}\zeta(3)\log^2 2-\frac{1}{12}\log^5 2\\
&-\frac{1}{8}\zeta(3)\log^2 2+\frac{1}{6}\log^5 2-\frac{1}{6}\log^5 2\\
=&-2\mathcal{S}+2\mathrm{Li}_5\left(\frac{1}{2}\right)+4\zeta(5)-\frac{\pi^4}{360}\log 2+\frac{1}{8}\zeta(3)\log^2 2-\frac{\pi^2}{3}\zeta(3)+\frac{1}{12}\log^5 2
\end{aligned}$$

For $\mathcal{I}_1$,

$$\begin{aligned}
\mathcal{I}_1 =&\frac{1}{6}\int_0^{\frac{1}{2}}\frac{\log^3 x\log(1-x)}{1-x}\mathrm{d}x\\
=&-\frac{1}{6}\sum_{n=1}^{\infty}H_n\frac{\partial^3}{\partial n^3}\left(\frac{1}{(n+1)2^{n+1}}\right)\\
=&\sum_{n=1}^{\infty}\frac{H_n}{(n+1)^4 2^{n+1}}+\sum_{n=1}^{\infty}\frac{\log 2H_n}{(n+1)^3 2^{n+1}}+\sum_{n=1}^{\infty}\frac{\log^2 2H_n}{2(n+1)^2 2^{n+1}}+\sum_{n=1}^{\infty}\frac{\log^3 2H_n}{6(n+1)2^{n+1}}\\
=&\mathcal{S}-\mathrm{Li}_5\left(\frac{1}{2}\right)+\frac{\pi^4}{720}\log 2-\frac{1}{16}\zeta(3)\log^2 2+\frac{1}{24}\log^5 2
\end{aligned}$$

Combining these four integrals gives

$$\begin{aligned}
\sum_{n=1}^{\infty}\frac{(-1)^n H_n}{n^4} =&3\mathcal{S}-6\mathrm{Li}_5\left(\frac{1}{2}\right)-\frac{31}{16}\zeta(5)-3\mathrm{Li}_4\left(\frac{1}{2}\right)\log 2+\frac{\pi^4}{240}\log 2\\
&-\frac{3}{2}\zeta(3)\log^2 2+\frac{\pi^2}{3}\zeta(3)+\frac{\pi^2}{12}\log^3 2-\frac{3}{40}\log^5 2
\end{aligned}$$

But consider $f(z)=\dfrac{\pi\csc(\pi z)(\gamma+\psi(-z))}{z^4}$.
At the positive integers,

$$\sum_{n=1}^{\infty}\mathrm{Res}(f,n)=\sum_{n=1}^{\infty}\operatorname*{Res}_{z=n}\left(\frac{(-1)^n}{z^4(z-n)^2}+\frac{(-1)^n H_n}{z^4(z-n)}\right)=\sum_{n=1}^{\infty}\frac{(-1)^n H_n}{n^4}+\frac{15}{4}\zeta(5)$$

At $z=0$,

$$\begin{aligned}
\mathrm{Res}(f,0)&=z^3\left(\frac{1}{z}+\frac{\pi^2}{6}z+\frac{7\pi^4}{360}z^3\right)\left(\frac{1}{z}-\frac{\pi^2}{6}z-\zeta(3)z^2-\frac{\pi^4}{90}z^3-\zeta(5)z^4\right)\\
&=-\zeta(5)-\frac{\pi^2}{6}\zeta(3)
\end{aligned}$$

At the negative integers,

$$\sum_{n=1}^{\infty} \operatorname{Res}(f, -n) = \sum_{n=1}^{\infty} \frac{(-1)^n H_n}{n^4} + \frac{15}{16}\zeta(5)$$

Since the sum of the residues is zero,

$$\sum_{n=1}^{\infty} \frac{(-1)^n H_n}{n^4} = -\frac{59}{32}\zeta(5) + \frac{\pi^2}{12}\zeta(3)$$

This implies that

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{H_n}{n^4 2^n} =& 2\mathrm{Li}_5\left(\frac{1}{2}\right) + \frac{1}{32}\zeta(5) + \mathrm{Li}_4\left(\frac{1}{2}\right)\log 2 - \frac{\pi^4}{720}\log 2 + \frac{1}{2}\zeta(3)\log^2 2 \\ & -\frac{\pi^2}{12}\zeta(3) - \frac{\pi^2}{36}\log^3 2 + \frac{1}{40}\log^5 2\end{aligned}$$